

AI-900: AI Fundamentals Trainer Preparation Guide

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Purpose

This course introduces fundamentals concepts related to artificial intelligence (AI), and the services in Microsoft Azure that can be used to create AI solutions. The course is not designed to teach students to become professional data scientists or software developers, but rather to build awareness of common AI workloads and the ability to identify Azure services to support them.

Learners will be able to identify and describe core AI concepts such as workloads and core services for AI and machine learning on Microsoft Azure.

This course can be taken as an optional first step in learning about cloud AI services with Microsoft Azure, before taking further Azure role-based certifications like Azure Data Scientist Associate or Azure AI Engineer Associate.

This course is designed to be delivered in one full day.

The course is designed as a blended learning experience that combines instructor-led training with online materials on the Microsoft Learn platform (<https://docs.microsoft.com/learn>). The hands-on exercises in the course are completed in the associated Microsoft Learn modules, and students are encouraged to use the content on Learn as reference materials to reinforce what they learn in class and to explore topics in more depth.

Audience Profile

The Azure AI Fundamentals course is designed for anyone interested in learning about the types of solution artificial intelligence (AI) makes possible, and the services on Microsoft Azure that you can use to create them. You don't need to have any experience of using Microsoft Azure before taking this course, but a basic level of familiarity with computer technology and the Internet is assumed.

Some of the concepts covered in the course require a basic understanding of mathematics, such as the ability to interpret charts. The course includes hands-on activities that involve working with data and running code, so a knowledge of fundamental programming principles will be helpful.

Audience Prerequisites

Prerequisite certification is not required before taking this course. Successful Azure AI Fundamental students start with some basic awareness of computing and internet concepts, and an interest in using Azure AI services.

Specifically:

- Experience using computers and the internet.
- Interest in use cases for AI applications and machine learning models.
- A willingness to learn through hands-on exploration.

Certification Exam

The study areas for the certification exam related to this course were last updated April 2022. Each study area has a percentage indicating the relative weight of the area on the exam. The higher the percentage, the more questions a candidate is likely to see in that area.

Study Area	Percentage
Describe Artificial Intelligence workloads and considerations	20-25%
Describe fundamental principles of machine learning on Azure	25-30%
Describe features of computer vision workloads on Azure	15-20%
Describe features of natural language processing workloads on Azure	25-30%
Describe Artificial Intelligence workloads and considerations	20-25%

Candidates should be able to identify and describe Azure AI and machine learning services. Practice tests for AI-900 are available [here](#).

Prerequisite Knowledge to teach this course

To successfully teach this course, instructors must have the following knowledge and skills:

- Familiarity with machine learning and AI software development.
- Programming skills – preferably with Python.
- Experience working with Azure Machine Learning.
- Experience developing software with Azure Cognitive Services.
- A working knowledge of the Azure portal and PowerShell.

Note: These guidelines are not inclusive of the requirements to become a Microsoft Certified Trainer (MCT).

Required Materials to prepare for and teach this course

You need the following materials to prepare for and teach this course:

Resource	Description
Microsoft PowerPoint files	Download the AI-900T00A-ENU-PowerPoint.zip from the MCT Download Center .
Change Log	Download the AI-900T00A-ENU-Change-Log.pdf from the MCT Download Center .
Lab environment provided by your lab hosting provider	Contact your lab hosting provider for instructions on using their lab environment.
Lab instructions	The lab instructions are provided in the lab environment and in the AI-900 Microsoft Learning GitHub repository https://github.com/microsoftlearning/ai-900-aifundamentals .
Student training content	See the following section for a detailed breakdown of each Learning Path covered in the course.
Microsoft Learn for Educators program	For educational institutions that deliver this course, additional assets are available through the Learning Download Center (LDC) for Educators.

Additional resources:

- [Microsoft Certified: Azure AI Fundamentals - Certifications | Microsoft Learn](#)
- [AI-900 Exam Study Guide \(microsoft.com\)](#)

Student training content

The student training content for this course is located in Microsoft Learn. The following table provides a breakdown of each Learning Path, the modules covered in each, and the link to each LP in Microsoft Learn.

The student training content includes links to additional reading material to help you prepare for

Learning Path	Module	Online training in Microsoft Learn
Course Introduction	N/A	Slides only
Learning Path 1: Introduction to AI	Module 1: Introduction to Artificial Intelligence Module 2: Artificial Intelligence in Microsoft Azure Lab: Explore Cognitive Services	Microsoft Azure AI Fundamentals: Get started with artificial intelligence - Training Microsoft Learn
Learning Path 2: Machine Learning	Module 1: Introduction to Machine Learning Module 2: Azure Machine Learning Lab: Explore Automated Machine Learning in Azure ML	Microsoft Azure AI Fundamentals: Explore visual tools for machine learning - Training Microsoft Learn
Learning Path 3: Computer Vision	Module 1: Computer Vision Concepts Module 2: Computer Vision solutions in Azure Lab: Explore Computer Vision	Microsoft Azure AI Fundamentals: Explore computer vision - Training Microsoft Learn
Learning Path 4: Natural Language Processing	Module 1: Introduction to Natural Language Processing Module 2: Building Natural Language Solutions in Azure Lab: Explore text analytics	Microsoft Azure AI Fundamentals: Explore natural language processing - Training Microsoft Learn

Additional Optional Content:

- [Microsoft Azure AI Fundamentals: Explore decision support - Training | Microsoft Learn](#)
- [Microsoft Azure AI Fundamentals: Explore knowledge mining - Training | Microsoft Learn](#)

Preparation Tasks

Instructors should complete the following tasks to prepare for teaching this course:

- If you have previously taught this class, refer to the course's Change Log. It provides detailed information on how the course has changed over time. The Change Log is updated for each course release.
- Review all topics in the student training material in Microsoft Learn (see the link in the Required Materials section above). You should be well-versed in every topic. If you have previously taught the course and are comfortable with your knowledge of each topic, focus primarily on the new or updated topics as outlined in the Change Log.

- Review the PowerPoint slides.
 - Be able to speak to each of the talking points on the slides. Some slides include a graphic from the associated Learn content for the topic. These graphics are provided on the slide so that you can speak to them to help explain the key talking points in the topic.
 - **The bulleted items on each slide should NOT be read verbatim to the students.** The students can read the slides themselves. Rather, the bullet points reflect the key information that you should focus on when discussing each topic. You should use your experience as a subject matter expert to explain the What, the Why, and the How of each topic. **This is your opportunity to provide a real value-add above and beyond the bulleted talking points.**
- Review the Additional Reading links and other linked resources provided in the student training material. **It's recommended that you present key points from this material to supplement the value-add you provide as an instructor.**
- As you prepare for the class, you should review each unit and determine which ones you want to perform demonstrations of the corresponding product functionality. It's up to you to decide which product features you want to demonstrate to the class. You should use your experience to identify key points during the demonstration process. **This is an area where you should rely on your experience as a subject matter expert to provide additional value-add to the students.**
- You should review each Knowledge Check (KC) question so that you know why the correct answer is correct for each question. Students may challenge some of the questions, so you must be able to address any of those concerns.
- You should perform the labs yourself prior to class so that you become familiar with them and with any of the difficult points in the lab exercises. This will prepare you for helping students in case they get stuck. Cloud-based services are subject to frequent changes, and can be affected by network configuration and transient network issues. If something goes wrong during a lab, avoid spending too long trying to troubleshoot it. If you can get the exercise working on your own computer, demonstrate it and encourage students to go back to the relevant module on Microsoft Learn later.
- Review some of the customer case studies at <https://www.microsoft.com/ai>. You can use these to show students how AI is being used in real scenarios. In particular, take a look at <https://www.microsoft.com/ai/ai-for-good> to focus on how AI makes it possible to create solutions that benefit society and the planet.
- Review the content at <https://www.microsoft.com/ai/responsible-ai> to become familiar with Microsoft's approach to responsible AI, and the 6 principles for AI developers to follow.

Course Timing

Daily Agenda

The following agenda provides estimated times to complete each classroom activity. However, the estimated times may vary depending on the background of your students, which may affect whether you can move faster or slower through the course material.

Estimated times for each Module include the time to complete:

- The module's PowerPoint slide deck presentation.
- Any pre-defined product demonstrations.
- Time to review Knowledge Check questions (see the section on Additional Timing Notes below).

- Time to complete a classroom discussion activity if a Discussion slide is included in the module slide deck.

You should adjust the agenda accordingly based on any classroom activities that you personally created or plan to deliver that are not included in the slides for this course. For example, if you plan to present:

- ad-hoc demonstrations
- review activities
- classroom games
- and so on...

Note: Each Learning Path/Module activity in the following agenda is the slide deck presentation for that module.

Day	Estimated Time	Classroom activity
1	10 minutes	Course Introduction (time may vary due to the number of student introductions in a given course)
	25 minutes	Learning Path 1, Module 1
	25 minutes	Learning Path 1, Module 2
	30 minutes	Introduction to AI Lab
	25 minutes	Learning Path 2, Module 1
	20 minutes	Learning Path 2, Module 2
	1 hour	Machine Learning Lab
	30 minutes	Learning Path 3, Module 1
	30 minutes	Learning Path 3, Module 2
	30 minutes	Computer Vision Lab
	30 minutes	Learning Path 4, Module 1
	30 minutes	Learning Path 4, Module 2
	30 minutes	Natural Language Processing Lab
	15 minutes	Summary and Next Steps

Additional Timing Notes - Knowledge Check questions

Knowledge check (KC) questions are provided throughout the course to check the student's knowledge of the material that was covered. Instructors can use these KC questions in several ways:

- Conduct a formal classroom exercise in which you go through the questions in a module before moving on to the next module.
- Sprinkle the questions into the content as you cover the related material for a module
- Let the students review the questions after class as a daily homework assignment. You can set aside time at the start of each day to answer any questions they have regarding the prior day's questions. This may be the most feasible option given the tight time constraints that most classes work under.

It will be left up to each instructor to determine how they want to incorporate the KC questions into their class.

If you provide students with time to review the KC questions at the end of specific topics and at the end of each module, you should provide a couple of minutes per question, along with a few extra minutes per question to respond to student questions or challenges concerning certain questions they may not understand or whose answers they disagree with. This may add an extra 15 minutes to complete each module.

Additional Timing Notes – Classroom Discussions

This course does not include formal classroom discussion questions. However, incorporating a few discussion questions is encouraged. Spread discussion questions throughout modules. You should plan on 5-10 minutes for each discussion.

- Sample discussion questions:
 - Could you see organizations implementing this feature? Why or why not?"
 - What are your key takeaways from this Learning Path?
 - What are the key features discussed in this Learning Path that you'd like to learn more about? Do you have a use case idea?

Labs

The labs must be completed within the lab environment provided by your lab hosting provider. Detailed, step-by-step instructions are provided for each lab and presented as part of the UI experience within your lab environment.

At the time the courses were released, the lab instruction had been thoroughly tested and the lab steps were 100% accurate. However, given the nature of Microsoft's cloud products and the fact that Microsoft releases UI updates on a regular basis, it's possible that at some point in time, the UI for a given feature may change so that it no longer matches the lab instruction.

If students encounter lab steps that don't accurately reflect the UI, they'll have to work through the UI to determine what needs to be done. Typically, UI changes are quite subtle, so hopefully you don't find yourself in a situation where a feature was completely overhauled.

However, if you do run into major UI changes, challenge your students to work through it, and only offer help if they definitely need it. Product UI changes will be part of their daily life in today's cloud-centric world. As IT/Pros, they must learn how to work through such situations.

One thing Microsoft does ask of you is that if you run into situations such as this where lab instructions no longer match the corresponding UI, please document the issue in the course's GitHub repository. This will help Microsoft's World-Wide Learning team update the lab instructions to keep them as up to date as possible. For information on how to submit an issue, please see [GitHub User Guide for MCTs](#).

Potential lab issues

This section identifies lab exercises in which students have run into problems in the past due to the complexity of the lab and/or the students lack of prerequisite background knowledge.

Some of these issues stem from changes to the product. If any of the instructions appear outdated, please file a bug so we can address it.

Lab and Exercise	Description
Lab: Explore Cognitive Services	- Students can keep their resource for later Computer Vision and Natural Language Processing labs - Students don't need to know specific capabilities of the Anomaly Detector service; the focus of this lab is on seeing Cognitive Services in action.
Lab: Explore Automated Machine Learning in Azure ML	Keep in mind for Azure Machine Learning Automated machine learning: a ML job can take close to an hour to complete. Optional Azure Machine Learning Designer labs exist. - If components in the asset library cannot be found, try to adjust the search bar filter. - When students create real-time inference pipelines, the pipelines with custom data inputs will not have a Web Input component automatically added to the canvas. Students will need to manually add the web input component, which is described in the instructions.

Azure Subscriptions

To complete the labs and any additional demonstration exercises in this course, students will need an Azure Subscription. If you are not using a hosted lab environment, you will be provided with Azure passes for you and your student. Additional information can be found on [Access to Microsoft Learning Azure Passes for Students of Authorized Microsoft Learning Partners](#). Please ensure that your Learning partner has requested Azure passes for both you and the students. After receiving the passes, each student will need to activate their pass.

The Azure pass effectively functions in the same way as the [publicly available Microsoft Azure Trial Subscription](#). This means there are limitations on what you can do with the pass.

Portal, Cloud Shell, PowerShell, and the CLI

The lab instructions are written to use the Cloud Shell. The Cloud Shell automatically connects to Azure and provides access to PowerShell and the CLI.

If you would rather have students use PowerShell or the CLI locally, you can use these links.

- [Install Azure PowerShell on Windows with PowerShellGet](#)
- [Install Azure CLI on Windows](#)

Tips and Tricks for teaching AI-900

Are you looking to improve when teaching this course? Maybe you don't teach the course very often, or you haven't taught the course before? If so, this section offers some tips from our top trainers to help you deliver the best course possible.

Here are three specific areas we would like to highlight: **Course timing**, **Hands-on labs**, and **Classroom interactivity**.

Course timing

Typical Comments:

Course timing is not a good fit, day ran too long/short, course should be an additional day, and not enough time to absorb the material.

Discussion:

Select the combination of course elements that fit best within the allotted classroom time. Consider your audience, consider your ability to tell the story, and consider the depth of coverage.

Here is how two different instructors presented the material. Notice they do not follow the prescribed course sequence and have selected content based on their teaching style.

Persona	Course timing approach
Instructor 1	I use the PPT slides during a course as a guideline and hide slides that I will whiteboard or demo. I use a lot of demonstrations and often include simple portal walk-throughs I've created myself. And I do a lot of whiteboarding to cover concepts and flowcharts. I survey the class and spend more time on product areas where the students are most interested.
Instructor 2	I start each lesson with a description of an AI or ML task and then show the steps in the portal. As I go through the demo, I describe the content covered in the slides. After the demo I quickly skimmed the slides, making sure everything was covered and there weren't any questions. I use the Summary slide to point students to other study materials available on Learn.

Summary:

To improve in this area, we ask that you fit the course to your audience and teaching style. Plan your time and set the students' expectations of what will be covered and how it will be covered.

Classroom interactivity

Typical comments: *Need more balance between lecture and hands-on, more discussion and question/answer, and lecture was too long.*

Discussion:

Our goal in Microsoft Learning is to ensure there is at least a 50/50 mix of lecture/interactive elements, such as demonstrations, labs, and review questions.

Here are some suggestions from our top trainers on how they incorporate interactivity in their classes.

Persona	Class interactivity approach
Instructor 1	• Frequently stop and ask open-ended questions. • Use Mentimeter or Office to display multiple-choice questions and review the student responses. • Let one of your advanced students share their screen and “drive” during the demonstration. Coach the student through the steps. • Have someone monitor the chat and let you know if there is a question.
Instructor 2	• Use a poll (or other signal) to determine if students are interested in or using a product or feature. • Ask how many students are planning to take the exam. Work that information into your presentation. • Ensure students can annotate your whiteboard. Have them participate by placing resources or linking items.

Summary:

To improve in this area, we ask that you plan to limit the amount of lecture to less than half the class time. Supplement your lecture with demos, questions, and discussion. Try to engage the students in as many ways as possible. This interactivity will take longer, so be sure to account for that.

Real-world use cases

Here are a few use cases provided by Microsoft. Trainers and students are encouraged to share their own use case experiences.

[Microsoft Customer Story-Uber boosts platform security with the Face API, part of Microsoft Cognitive Services](#)

[Microsoft Customer Story-Machine translation speaks Volkswagen – in 40 languages](#)

[Microsoft Customer Story-KPMG helps financial institutions save millions in compliance costs with Azure Cognitive Services](#)

Feedback

In this course, we have provided a framework for you to work with. Take time to prepare and think about the value that only an instructor can bring to training. We hope to partner with you to provide an exceptional student experience, and we welcome your feedback.